Shadow Analytics

AI-Powered Document Intelligence Platform

View project: https://lnkd.in/p/gP2C4hG5

THE PROBLEM

1. Businesses spend a lot of time reading PDFs, scanned documents, and reports manually.
2. Finding specific information in these files is slow and repetitive.
3. Manual work can lead to mistakes and missed information.
4. As the number of documents grows, it becomes harder to process them quickly and make decisions.

MY APPROACH

Instead of building another generic document reader, I designed the system around the actual decision users need to make. Users can upload documents, define the information they need, and let an AI pipeline extract, validate and structure the required information.

WHAT I BUILT

- Built an end-to-end AI document intelligence platform using Python, OCR/Tesseract, Gemini/LLMs, RAG and structured extraction pipelines.
- Designed the workflow: Upload → OCR → Extraction → Validation → Confidence Scoring → Human Review → Structured Output.
- Added custom questions/fields, allowing users to define what information the AI should extract.
- Added confidence scores and bounding boxes so users can understand and verify where extracted information came from.
- Built filtering, custom columns and Excel export for downstream analysis.
- Designed the platform as a reusable SaaS architecture so the same workflow can support different document types and industries.

IMPACT

> 90–98% extraction accuracy and approximately 85% reduction in manual document-processing effort, turning large volumes of unstructured documents into structured, decision-ready information.